

12. Test Results and Observations

12.1 Overview

After executing all API requests using both **Postman Collection Runner** and **Newman Command Line Execution**, the test results were analyzed to verify whether the ReqRes APIs behaved according to the expected requirements.

The purpose of this phase was to:

- Verify successful execution of all APIs
- Analyze test validation results
- Identify any failures or issues
- Document observations for reporting purposes

The execution confirmed that the APIs responded correctly and all validation checks were successfully completed.

12.2 Test Execution Summary

The following APIs were executed during testing:

Sr. No.	API Name	Method	Execution Status
1	Login API	POST	Passed
2	Get Users API	GET	Passed
3	Get Single User API	GET	Passed
4	Create User API	POST	Passed
5	PUT Update User API	PUT	Passed
6	PATCH Update User API	PATCH	Passed
7	Delete User API	DELETE	Passed

Overall Result

Total APIs Executed : 7
Passed : 7
Failed : 0
Success Rate : 100%

12.3 Login API Results

Test Objective

Verify successful user authentication and token generation.

Expected Result

- Status Code = 200
- Token should be generated

Actual Result

- Status Code = 200
- Token received successfully

Observation

The Login API authenticated the user successfully and returned a valid token in the response body.

Status

✓ Passed

12.4 Get Users API Results

Test Objective

Retrieve list of users from the system.

Expected Result

- Status Code = 200
- User list returned

Actual Result

- Status Code = 200
- User data received successfully

Observation

The API returned paginated user information with all required fields.

Status

✓ Passed

12.5 Get Single User API Results

Test Objective

Retrieve details of a specific user.

Expected Result

- Status Code = 200
- User information available

Actual Result

- Status Code = 200
- User details returned successfully

Observation

The API correctly returned user information associated with the specified user ID.

Status

✓ Passed

12.6 Create User API Results

Test Objective

Create a new user.

Expected Result

- Status Code = 201
- User ID generated

Actual Result

- Status Code = 201
- User created successfully

Observation

The API generated a new user record and returned the user ID along with creation timestamp.

Status

✓ Passed

12.7 PUT Update User API Results

Test Objective

Perform complete update of user information.

Expected Result

- Status Code = 200
- Updated values returned

Actual Result

- Status Code = 200
- User information updated successfully

Observation

The API successfully updated all supplied fields and returned the updated information.

Status

✓ Passed

12.8 PATCH Update User API Results

Test Objective

Perform partial update of user information.

Expected Result

- Status Code = 200
- Modified fields updated

Actual Result

- Status Code = 200

- Partial update successful

Observation

Only the specified fields were updated while the request completed successfully.

Status

✓ Passed

12.9 Delete User API Results

Test Objective

Delete an existing user.

Expected Result

- Status Code = 204
- No response body

Actual Result

- Status Code = 204
- User deleted successfully

Observation

The API processed the deletion request successfully and returned the expected response.

Status

✓ Passed

12.10 Negative Testing Results

Scenario Tested

Invalid User ID

Request

GET {{base_url}}/users/23

Expected Result

404 Not Found

Actual Result

404 Not Found

Observation

The API handled invalid requests correctly by returning the appropriate error status code.

Status

✔ Passed

12.11 Validation Results Summary

The following validations were performed across all APIs:

Validation Type	Result
Status Code Validation	Passed
Response Body Validation	Passed
Response Time Validation	Passed
JSON Schema Validation	Passed
Negative Testing Validation	Passed

Observation

All validation scripts executed successfully without any assertion failures.

12.12 Performance Observations

Response times were monitored during execution.

Findings

- All APIs responded within acceptable limits.
- No significant delays were observed.
- Response times remained consistent across executions.

Average Response Time

API	Average Response Time
Login API	< 500 ms
Get Users API	< 500 ms
Get Single User API	< 500 ms
Create User API	< 500 ms
Update APIs	< 500 ms
Delete API	< 500 ms

Observation

The ReqRes APIs demonstrated stable and reliable performance during testing.

12.13 Collection Runner Results

Execution Outcome

```
Requests Executed : 7
Assertions Passed : 20
Assertions Failed : 0
```

Observation

All requests and validations executed successfully through Postman Collection Runner.

Status

✓ Successful

12.14 Newman Execution Results

Execution Outcome

```
Iterations : 1
Requests   : 7
Assertions : 20
```

```
Passed      : 20
Failed      : 0
```

Observation

The Newman execution completed successfully and produced the expected test results without errors.

Status

✔ Successful

12.15 Key Observations

The following observations were made during testing:

- 1. All API endpoints were accessible and operational.
- 2. Expected HTTP status codes were returned for every request.
- 3. Response bodies contained the required data fields.
- 4. Negative test cases returned proper error responses.
- 5. Response times remained within acceptable limits.
- 6. JSON response structures were consistent.
- 7. Collection Runner execution completed successfully.
- 8. Newman execution completed successfully.
- 9. No critical defects were identified.
- 10. Overall API behavior was stable and reliable.

12.16 Final Test Result Summary

Parameter	Result
Total APIs Tested	7
Total APIs Passed	7
Total APIs Failed	0
Total Assertions Passed	20
Total Assertions Failed	0
Success Rate	100%

12.17 Conclusion

The execution results indicate that all ReqRes API endpoints functioned as expected. All requests passed their validation checks, including status code verification, response body validation, response time checks, and JSON schema verification. Both Postman Collection Runner and Newman executions completed successfully without any failures.

Based on the test execution results, the ReqRes APIs are considered stable, reliable, and suitable for demonstrating API testing concepts using Postman and Newman.